Concepts of Probability

Statistical Foundations of AI and ML, Module 1

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Contents

Background

Outcomes and Events

- Ω: Outcome space or Sample space of a random experiment (set of all possible outcomes)
- Eg. in a coin-toss experiment: $\Omega = \{H, T\}$
- Eg. in a 3-coin-toss experiment: $\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$
- An event: a set of possible outcomes
- Eg. in a 3-coin-toss experiment, we get 1 head only $\{HTT, THT, TTH\}$
- $\mathcal{A}$: Set of *events*, a subset of the power-set of Ω
- Practically, outcomes are raw data but events are what we really care about
- $\mathcal{B}$: a σ-algebra on $\mathcal{A}$

Events and Measurable Space

- A σ-algebra $\mathcal{B}$ on Ω is a set of subsets of Ω
- It must satisfy the following criteria
  1. $\Omega \in \mathcal{B}$
  2. $A \in \mathcal{B} \rightarrow A^C \in \mathcal{B}$ where A is any subset of Ω (closed under complement)
  3. $A_1, \ldots, A_i, \cdots \in \mathcal{B} \rightarrow \cup_{i=1} A_i \in \mathcal{B}$ (closed under countable union)
- A σ-algebra $\mathcal{B}$ can be *generated* from an event set $\mathcal{A}$ by adding new subsets of Ω to satisfy above criteria
- eg. in a coin-toss experiment, $\mathcal{A} = \{\{H\}, \{T\}\}$, $\mathcal{B} = \{\phi, \{H\}, \{T\}, \{\Omega\}\}$
- $(\Omega, \mathcal{B})$: a *measurable space*
- Define a *probability measure* on elements of $\mathcal{B}$, and we get a *probability space* $(\Omega, \mathcal{B}, P)$!

Probability Measure and Space

- Axiomatic Definition of Probability in terms of $(\Omega, \mathcal{B}, P)$
- Probability measure $P : \mathcal{B} \rightarrow [0, 1]$ such that
  1. $P(\Omega) = 1$ (remember: $\Omega \in \mathcal{B}$)
  2. $P(A) = 1 - P(A^C)$ for any $\cancel{B \in \Omega}$ $A \in \mathcal{B}$
  3. $P(\cup_i A_i) = \sum_{i=1} P(A_i)$ where $A_1, \ldots, A_i, \cdots \in \mathcal{B}$ and are **disjoint** subsets of Ω
- Eg. in a coin-toss experiment, $P(\Omega) = 1, P(\{H\}) = 0.6, P(\{T\}) = 0.4, P(\phi) = 0$
- Conditional probability $P(A|B) = \frac{P(A \cap B)}{P(B)}$ where $A, B \in \mathcal{B}$ (any two events)
- Two events A,B are disjoint iff $P(A \cap B) = P(A) * P(B)$
- Bayesian Theorem: $P(A|B) * P(B) = P(B|A) * P(A)$

Random Variables

- A random variable defines a mapping $X : \Omega \rightarrow \mathcal{R}$
- $X^{-1}((-\infty, x]) = \{\omega : X(\omega) \leq x\}$, and $X^{-1}((-\infty, x]) \in \mathcal{B}$ $\forall x$
- Each outcome is associated with a real number
- Also, X inverse-maps each real interval $(-\infty, x)$ to an element of $\mathcal{B}$
- In case of coin-toss, $X(\{H\}) = 1$, $X(\{T\}) = 0$ by convention (other numbers could also be used, but this choice makes mathematical expressions more elegant)
- Range of X finite/countable: X is Discrete Random Variable (DRV)
- Range of X uncountable: X is Discrete Random Variable (CRV)

Distribution Functions

- Support of X: set of unique value of X (finite/countable for DRV, not for CRV)
- Cumulative Distribution Function (CDF): $F_X(x) = P(X \leq x)$, i.e. probability of the event formed by outcomes for which X is at most x
- Probability Mass Function (PMF) for DRVs: $p_X : Supp(X) \to [0,1]$ and $p_X(x) = P(X = x)$
- $Supp(X) = \{x_1, x_2, \ldots, x_n\}$, $0 \leq p_X(x_i) \leq 1 \forall i$ and $\sum_{i=1}^{n} p_X(x_i) = 1$
- DRV: $F_X(x) = \sum_{i=1}^{n} p_X(x_i) I(x_i \leq x)$ where $Supp(X) = \{x_1, x_2, \ldots, x_n\}$
- CRV: $P(X = x) = 0 \forall x \in \mathcal{R}$, i.e. point-mass is 0 everywhere
- CRV: Probability Density Function PDF $f_X(x) = F'_X(x) = P(x - dx < X < x + dx)$

Joint and Conditional Distributions

- X, Y any two random variables with PMF p_X, p_Y
- Joint distribution
  $p_{X,Y}(x,y) = P(X = x, Y = y) = P(\omega | X(\omega) = x, Y(\omega) = y)$
- Conditional distribution $p_{X|Y}(x,y) = P(X = x | Y = y) = \frac{P(X-x, Y=y)}{P(Y=y)}$
- Can be analogously extended for continuous distributions and PDFs
- X, Y independent:
  $p_{X,Y}(x,y) = p_X(x) p_Y(y) \forall x \in \{Supp(X)\}, y \in \{Supp(Y)\}$
- Mutual Information
  $I(X,Y) = \sum_{x \in Supp(X), y \in Supp(Y)} P_{X,Y}(x,y) log(\frac{P_{X,Y}(x,y)}{P_X(x) P_Y(y)})$
- Law of total probability: $p_X(x) = \sum_{y \in Supp(Y)} p_{X,Y}(x,y)$ (DRV),
  $p_X(x) = \int_{y \in Supp(Y)} p_{X,Y}(x,y) dy$ (CRV)

Common Probability Distributions

Common Discrete Distributions

Distribution	Support	PMF	Parameters
Bernoulli	$\{0,1\}$	$p^x(1-p)^{(1-x)}$	$0 \leq p \leq 1$
Binomial	$\mathcal{Z}$	$\binom{N}{x}p^x(1-p)^{N-x}$	$N \in \mathcal{Z}^+, 0 \leq p \leq 1$
Poisson	$\mathcal{Z}$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$\lambda \in \mathcal{R}^+$
Geometric	$\mathcal{Z}^+$	$(1-p)^{x-1}p$	$0 \leq p \leq 1$
Categorical	$\{V_1, \ldots, V_K\}$	$\prod_{k=1}^{K} p_k^{I(x=V_k)}$	$0 \leq (p_1, p_2, \ldots, p_K) \leq 1$ $\sum_{i=1}^{k} p_i = 1$
Multinomial	$\mathcal{Z}^K$	$\frac{N!}{x_1!..x_K!}\prod_{k=1}^{K} p_k^{x_k}$	$N \in \mathcal{Z}^+, \sum_{i=1}^{k} p_i = 1$ $0 \leq (p_1, \ldots, p_K) \leq 1$

Common Continuous Distributions

Distribution	Support	PDF	Parameters
Uniform	$\mathcal{R}$	$\frac{1}{b-a}$	$(a,b) \in \mathcal{R}$
Beta	$(0,1)$	$\frac{1}{B(a,b)} x^{(a-1)}(1-x)^{(b-1)}$	$(a,b) \in \mathcal{Z}^+$
Exponential	$\mathcal{R}^+$	$\lambda exp(-\lambda x)$	$\lambda \in \mathcal{R}+$
Gamma	$\mathcal{R}+$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} exp(-\beta x)$	$(\alpha,\beta) \in \mathcal{R}^+$
Gaussian	$\mathcal{R}$	$\frac{1}{\sqrt{(2\pi)}\sigma} exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$	$\mu \in \mathcal{R}, \sigma \in \mathcal{R}^+$
χ^2	$\mathcal{R}^+$	$\frac{x^{K/2-1} e^{-x/2}}{2^{\frac{k}{2}} \Gamma(K/2)}$	$K \in \mathcal{Z}^+$

Common Continuous Distributions

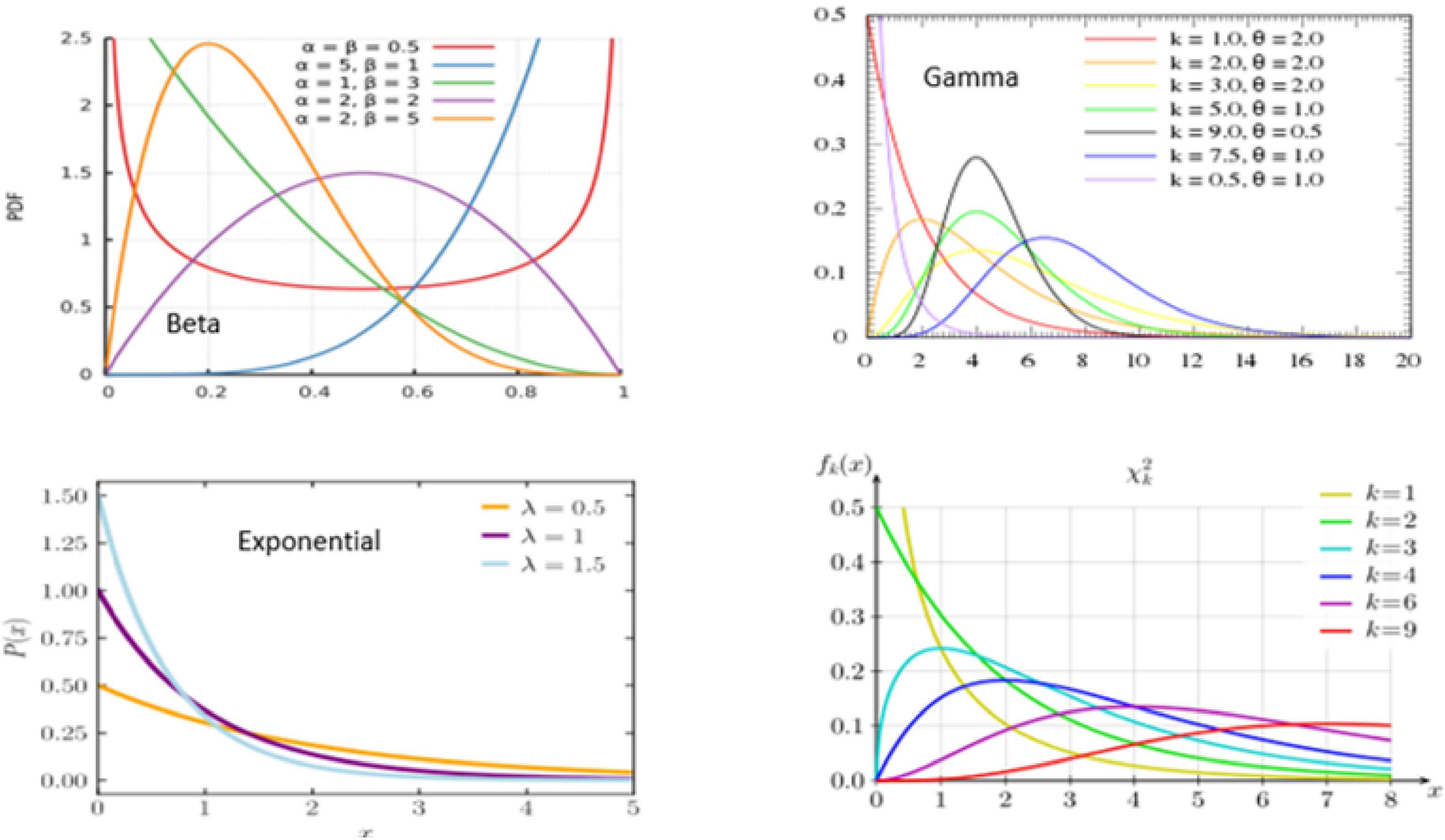

Figure: PDFs of continuous distributions for different parameter settings

Additional Distributions

- If $Z_i \sim \mathcal{N}(0,1)$, then $\sum_{i=1}^{k} Z_i^2 \sim \chi_k^2$
- $\chi^2(k)$ is equivalent to $\Gamma(k/2, 1/2)$
- If $Z \sim \mathcal{N}(0,1)$ and $Y \sim \chi_k^2$, then $\frac{Z}{\sqrt{(Y/k)}}$ follows t_k-distribution
- If $Y_1 \sim \chi_{k1}^2$ and $Y_2 \sim \chi_{k2}^2$, then $\frac{Y1/k1}{Y2/k2} \sim F_{k1,k2}$
- **HOMEWORK**: find their PDFs!!
- **Key Property**: $X \sim \mathcal{N}(\mu, \sigma^2) \rightarrow \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$
- **Key Property**: Gaussian CDF $\Phi(x) = \frac{1}{\sqrt{(2\pi)}\sigma} \int_{-\infty}^{x} exp(-\frac{(x-\mu)^2}{2\sigma^2})dx$
- **Key Property**: $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \rightarrow X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
- **Key Property**: $X_1 \sim Gamma(a_1, b), X_2 \sim Gamma(a_2, b) \rightarrow X_1 + X_2 \sim Gamma(a_1 + a_2, b)$

Additional Distributions

Distribution	Support	PDF	Parameters
Dirichlet	PMFs	$\frac{1}{B(\alpha_1,\ldots,\alpha_K)}\prod_{i=1}^{K} x_i^{\alpha_i-1}$	$\alpha_1,\ldots,\alpha_K \in \mathcal{R}$
M.V. Gaussian	$\mathcal{R}^D$	$\frac{exp(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))}{2\pi^{\frac{D}{2}}\lvert\Sigma\rvert^{\frac{1}{2}}}$	(μ,Σ)

- Every draw from a Dirichlet Distribution is valid P.M.F.
- Mean parameter μ of MV Gaussian is a vector in $\mathcal{R}^D$
- Parameter Σ of MV Gaussian is a Covariance Matrix: a $D \times D$ positive definite matrix

Further Properties

Expectations and Moments

- Expectation of a DRV $E(X) = \sum_{x \in Supp(X)} x p_X(x)$
- Expectation of a CRV $E(X) = \int_{x \in Supp(X)} x f_X(x) dx$
- Variance $V(X) = E((X - E(X))^2) = E(X^2) - (E(X))^2$
- t-th Moment: $E(X^t)$, t-th Central Moment: $E((X - E(X))^t)$
- $X \sim Ber(p) \rightarrow E(X) = p, V(X) = p(1-p)$
- $X \sim Binom(p) \rightarrow E(X) = Np, V(X) = Np(1-p)$
- $X \sim Poiss(\lambda) \rightarrow E(X) = \lambda, V(X) = \lambda$
- $X \sim \mathcal{N}(\mu, \sigma^2) \rightarrow E(X) = \mu, V(X) = \sigma^2$
- $X \sim Beta(a, b) \rightarrow E(X) = a/(a+b), V(X) = \frac{ab}{(a+b)^2(a+b+1)}$
- $X \sim Gamma(\alpha, \beta) \rightarrow E(X) = \alpha/\beta, V(X) = \alpha/\beta^2$

Moment Generating Functions

- M.G.F. of a RV X: $E(e^{tX})$
- Obtaining n-th moment from MGF: calculate n-th dervative of MGF and set $t = 0$
- Two random variables with same MGF are identically distributed
- This property allows us to calculate distributions of many additive functions of multiple independent RVs:
- Eg. MGF of $X_1 + X_2 = E(e^{t(X_1+X_2)}) = E(e^{tX_1})E(e^{tX_2})$
  1. Calculate the PDF of the sum of independent Gaussian RVs with different parameters
  2. Calculate PDF of $\chi^2(k)$ using MGF

Mean Variable and Central Limit Thorem

- Consider independent and identically distributed RVs $X_1, X_2, \ldots, X_n$
- Define $S_n = \sum_{i=1}^{n} X_i$ and $\bar{X} = \frac{S_n}{n}$
- Define $\mu = E(X_i)$ and $\sigma^2 = Var(X_i) < \infty$
- Define $T_n = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$
- Central Limit Theorem: As $n \to \infty$, $F_{T_n}(x) \to \Phi(x)$ (CDF of Standard Gaussian)
- i.e. T_n **converges in distribution** to $\mathcal{N}(0,1)$ as n approaches infinity
- **Key property**: If $X_i \sim \mathcal{N}(\mu, \sigma)$, then even for finite n, $T_n \sim \mathcal{N}(0,1)$
- **Key property**: If $X_i \sim \mathcal{N}(\mu, \sigma)$, then $\frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \bar{X})^2 \sim \chi^2(n-1)$

Multivariate Random Variables

- A vector/multivariate random variable $X : \Omega \to \mathcal{R}^D$
- Can be considered as a collection of D RVs $X_1, \ldots, X_D$, not necessarily independent!
- Expectation $E(X) \in \mathcal{R}^{\mathcal{D}}$, where $E(X)_i = E(X_i)$, i.e. each element in the expectation vector is the expectation of that RV
- Second moment $\Sigma = E((X - E(X))(X - E(X))^T)$, a $D \times D$ matrix called **Covariance Matrix**
- $\Sigma_{ij} = E(X_i - E(X_i))(X_j - E(X_j))$, called Covariance between X_i and X_j
- Diagonal elements Σ_{ii}: variance of each variable X_i
- Off-diagonal elements $\Sigma_{ij} = 0$ if X_i, X_j are independent

Comparing Distributions

- How to characterize or compare the distribution of any RV?
- Expectation of a function f w.r.t. distribution p: $E_p(f) = \sum_{x \in Supp(X)} (f(x)p(x))$ where p is a PMF (analogously for PDF)
- Entropy of a distribution p $h(p) = E_p(-log(p)) = -\sum_{x \in Supp(X)} (p_X(x)log(p_X(x)))$ where $p(X)$ is the PMF of X (analogously for PDF)
- Entropy is a measure of uncertainty, an RV with high probability mass concentration has low entropy
- Cross-entropy: used to compare two distributions p and q having same support
- Defined as $E_p(-log(q)) = -\sum_{x \in Supp(X)} (p(x)log(q(x)))$
- Low value indicates the two distributions are more similar mutual information, KL-divergence

Comparing Distributions

- Another important measure to compare two distributions with same support is the Kullback-Leibler Divergence
- $KL(p||q) = E_p(log(\frac{p}{q})) = \sum_{x \in Supp(X)} p(x) log(\frac{p(x)}{q(x)})$ (analogously for CRVs)
- Less value indicates higher similarity, equals 0 when $p = q$

The familiar Markov bounds

Proposition (Markov's inequality)

If $X \geq 0$*, then* $\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$ *for all* $t \geq 0$*.*

Proposition (Chebyshev's inequality)

For any $t \geq 0$*,* $\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\mathsf{Var}(X)}{t^2}$

Sub-gaussian random variables

A mean-zero random variable X is σ^2-*sub-Gaussian* if

$$\mathbb{E}[\exp(\lambda X)] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right) \quad \text{for all } \lambda \in \mathbb{R}.$$

(many equivalent definitions; see Vershynin or exercises)

Example

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]))] =$$

Example

If $X \in [a, b]$, then

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]))] \le \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Tensorization identities

- variance inequality familiar: if X_i are independent,

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i)$$

Proposition

If X_i are independent and σ_i^2-sub-Gaussian, then $\sum_{i=1}^n X_i$ is $\sum_{i=1}^n \sigma_i^2$ sub-Gaussian.

Chernoff and Hoeffding Inequalities

Corollary (Chernoff bounds for sub-Gaussian random variables)

Let X be σ^2-sub-Gaussian. Then

$$\mathbb{P}\left(X - \mathbb{E}[X] \geq t\right) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Corollary (Hoeffding bounds)

If X_i are independent σ_i^2-sub-Gaussian random variables,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i]) \geq t\right) \leq \exp\left(-\frac{nt^2}{\frac{2}{n}\sum_{i=1}^{n}\sigma_i^2}\right).$$

- usually stated as $X_i \in [a, b]$, so bound is $\exp(-\frac{2nt^2}{(b-a)^2})$

Maxima of sub-Gaussian random variables

- often want to control deviations of maximum (supremum in ULLNs)

Proposition

Let $\{Z_i\}_{i=1}^N$ be σ^2-sub-Gaussian (not necessarily independent). Then

$$\mathbb{E}\left[\max_i Z_i\right] \leq \sqrt{2\sigma^2 \log N}.$$

Sub-exponential random variables

- more nuanced control if variance small, or sub-gaussian parameter unavailable

Definition (Sub-exponential)

A random variable X is (τ^2, b)-sub-exponential if

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]))] \leq \exp\left(\frac{\lambda^2\tau^2}{2}\right) \quad \text{for } |\lambda| \leq \frac{1}{b}$$

Proposition (Tail bounds for sub-exponentials)

If X is (τ^2, b)-sub-exponential, then

$$\mathbb{P}\left(|X - \mathbb{E}[X]| \geq t\right) \leq 2\exp\left(-\min\left\{\frac{t^2}{2\tau^2}, \frac{t}{b}\right\}\right)$$